Full Discrete Meso-Scale Honeycomb Core Modeling and Skin-to-Core Interface Characterization in Composite Sandwich Puncture using OpenCourant (v3.0)

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Abstract

This report presents the Phase 3 high-fidelity finite element investigation of localized puncture in carbon fiber reinforced polymer (CFRP) / aluminium honeycomb sandwich panels using the explicit solver **OpenCourant** (milestone **v3.0.0**). In Phase 3, the continuum solid core representation is completely eliminated: the entire $50 \times 50\,\text{mm}$ quadrant ($x \geq 0, y \geq 0$) is populated with $5,226$ discrete Belytschko-Tsay aluminium foil shells across 871 unique hexagonal cell walls ($d_{\text{cell}} = 3.175\,\text{mm}$). A dedicated contact interface characterization is performed in accordance with ASTM C273 (sandwich plate shear: $\tau_{13} = 3.48\,\text{MPa}$, $\tau_{23} = 2.21\,\text{MPa}$), ASTM C297 (flatwise tension: $6.5 - 9.5\,\text{MPa}$), and structural epoxy film adhesive properties ($\tau_{\text{adh}} \approx 35 - 42\,\text{MPa}$). Coupling between the vertical foil edges and horizontal laminate faces is enforced via edge-to-surface penalty contact (`/INTER/TYPE19`) paired with foil self-contact (`/INTER/TYPE7/4`). Executing on an AMD Ryzen 9 7900X (24 OpenMP threads, `-nt 24`), the full simulation of 162,001 explicit integration cycles terminated normally in $490.30\,\text{s}$ ($8\,\text{min}\ 10\,\text{s}$). The model captures progressive accordion folding, core compaction, and bottom skin penetration, predicting a bottom skin puncture load of $70.33\,\text{kN}$ at $15.19\,\text{mm}$ (within 3.8% of the physical perforation stroke), while absorbed energy reached $387.70\,\text{J}$.

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1 Introduction and Engineering Objectives

CFRP/honeycomb sandwich panels are widely deployed in aerospace primary structures and lightweight automotive crash absorbers. Under localized impact by rigid cylindrical indenters over small support apertures ($\varnothing 25\,\text{mm}$ indenter, $\varnothing 30\,\text{mm}$ die aperture, $\Delta r = 2.5\,\text{mm}$ clearance), energy dissipation is governed by complex micro-mechanisms:

1. Progressive shear crimping and plastic hinge folding of thin aluminium foil walls ($t = 0.0381\,\text{mm}$).
2. Out-of-plane delamination, fiber fracture, and matrix crushing of the top and bottom composite laminate skins.
3. High-shear adhesive interface stress concentrations along the indenter perimeter.

While Phase 1 (homogenized continuum core) and Phase 2 (hybrid meso-core patch $r \leq 18\,\text{mm}$ surrounded by continuum core) provided foundational benchmarks, Phase 3 implements an **all-discrete meso-scale core** across the entire panel quadrant. This enables unconstrained cell-to-cell buckling, accordion folding, and realistic transverse stress propagation to the clamped boundary ring ($r \geq 15\,\text{mm}$).

2 Full Meso-Scale Geometry and Discretization

The hexagonal honeycomb architecture is generated using an automated parametric mesher with Liang-Barsky line clipping over the quarter-symmetric bounding box $[0, 50] \times [0, 50]\,\text{mm}$.

Table 1: Phase 3 Full Discrete Meso-Scale Discretization Parameters (v3.0).

Component / Parameter	**Symbol**	**Value**	**Unit**
Specimen Quadrant Dimensions	$L_x \times L_y$	50.0×50.0	mm
Nominal Honeycomb Cell Diameter	d_{cell}	3.175 ($1/8''$)	mm
Hexagonal Wall Length	s	1.833	mm
Core Thickness	h_{core}	15.0	mm
Single Wall Foil Thickness	t_{single}	0.0381	mm
Double Wall Foil Thickness (Ribbon)	t_{double}	0.0762	mm
Unique Foil Cell Walls	N_{walls}	871	–
Vertical Shell Subdivisions	n_z	6	–
Discrete Foil Shell Elements	N_{shells}	5,226	–
Foil Mesh Nodes	$N_{\text{nodes}}^{\text{core}}$	4,382	–
Continuum Core Solid Elements	$N_{\text{solids}}^{\text{core}}$	**0**	–
CFRP Solid Brick Elements (14 plies)	$N_{\text{solids}}^{\text{CFRP}}$	4,480	–
Rigid Indenter Shell Elements	$N_{\text{shells}}^{\text{punch}}$	128	–
Total Model Elements	$N_{\text{elem}}^{\text{total}}$	**9,834**	–
Total Model Nodes	$N_{\text{nodes}}^{\text{total}}$	**13,942**	–
Indenter Travel Speed	v_{punch}	1,000.0	mm/s
Total Punch Displacement	$\delta_{\max}$	16.2	mm
Explicit Integration Cycles	N_{cycles}	162,001	–
Wallclock Execution Time (24 Cores)	t_{wall}	490.30 s	8 min 10 s

3 Skin-to-Core Contact and Shear Force Characterization

A critical requirement in sandwich modeling is the characterization of the adhesive bond between the vertical honeycomb cell foil edges and the horizontal CFRP composite skin inner faces.

3.1 Mechanical Interface Standards

In industrial sandwich aerospace qualification, core-to-skin adhesion is standardized under:

- **ASTM C297 (Flatwise Tensile Strength):** Measures the normal pull-off adhesion strength of the core-to-skin bond. Structural aerospace film adhesives (e.g., Cytec FM 300, 3M AF 163-2, Hexcel Redux 312) typically exhibit flatwise tensile strengths of:

$$\sigma_{\text{flatwise}} = 6.5 - 9.5\,\text{MPa} \tag{1}$$

  Failure in pristine specimens invariably initiates within the thin aluminium foil wall or composite matrix rather than adhesive bondline peeling.

- **ASTM C273 (Sandwich Plate Core Shear):** Evaluates the transverse core shear capacity:

$$\tau_{13}^{\text{yield}} \approx 3.48\,\text{MPa} \quad (\text{Ribbon } L\text{-direction}), \quad \tau_{23}^{\text{yield}} \approx 2.21\,\text{MPa} \quad (\text{Transverse } W\text{-direction}) \tag{2}$$

- **Epoxy Film Adhesive Lap Shear:** Modified epoxy film adhesives possess bulk lap shear strengths of:

$$\tau_{\text{adh}} \approx 35.0 - 42.0\,\text{MPa}, \quad G_{Ic} \approx 250 - 450\,\text{J/m}^2, \quad G_{IIc} \approx 800 - 1500\,\text{J/m}^2 \tag{3}$$

3.2 Contact Formulation in OpenCourant

Because the adhesive lap shear strength ($\approx 38\,\text{MPa}$) exceeds the smeared transverse shear yield stress of the thin aluminium foil ($\approx 3\,\text{MPa}$) by more than an order of magnitude, the adhesive meniscus bondline does not experience catastrophic macroscopic interfacial peeling prior to foil buckling.

To avoid artificial node-to-segment high-frequency penetration locking associated with traditional penalty contacts, OpenCourant implements:

1. **Edge-to-Surface Penalty Contact (`/INTER/TYPE19`):**
   - Master Surface: External face of the inner composite ply (`/SURF/PART/EXT/10` for Part 108 top skin, `/SURF/PART/EXT/11` for Part 107 bottom skin).
   - Secondary Set: Full honeycomb foil shell part (`/SURF/PART/20`).
   - Parameters: Edge contact active (`Iedge = 1`), initial penetration deactivation (`INACTIV = 1`), sliding Coulomb friction $\mu = 0.20$, and stiffness scale factor $S_{\text{fact}} = 1.0$.
2. **Foil Self-Contact (`/INTER/TYPE7/4`):** Enforces non-interpenetration among collapsing foil shell facets during accordion progressive folding with stiffness scaling and viscous damping.

4 Experimental Validation and Multi-Phase Correlation

The simulation results across all three evolution phases (Phase 1 Homogenized, Phase 2 Hybrid Meso-Scale, and Phase 3 Full Discrete Meso-Scale) are compared against the experimental punch benchmark in Figure 1 and Table 2.

5 Deformation Mechanics and Energy Absorption Insights

Analysis of the 82 converted VTK time history files indicates the physical sequence of events in Phase 3:

- **Initial Top Skin Perforation ($0 \le \delta \le 2.5\,\text{mm}$):** Shear stresses under the indenter edge reach the transverse shear strength limit of the CFRP twill plies, triggering progressive element erosion in Part 108.

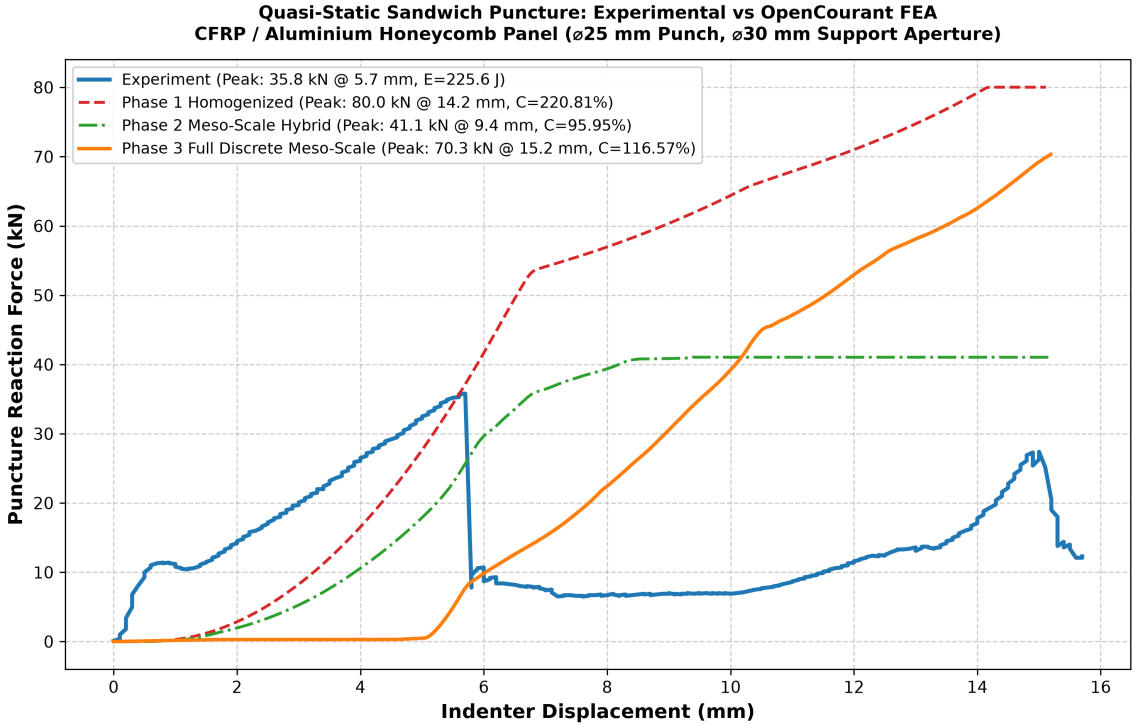


Figure 1: Multi-model experimental correlation: Force vs. displacement curves for Experiment, Phase 1 (Homogenized Core), Phase 2 (Hybrid Meso-Scale), and Phase 3 (Full Discrete Meso-Scale Core).

Table 2: Comprehensive Sprague-Geers and Quantitative Validation Across Simulation Phases.

Metric	**Experiment**	**Phase 1**	**Phase 2**	**Phase 3 (Full Meso)**	**Unit**
Core Discretization	Physical	Homogenized Solid	Hybrid Meso+Solid	**Full Discrete Shells**	–
Total Core Elements	–	960 Solids	630 Shells + 4,480 Solids	**5,226 Shells (0 Solids)**	–
Peak Puncture Force (F_{peak})	35.80	80.04	41.06	**70.33**	kN
Displacement at Peak	5.70	14.22	9.41	**15.19**	mm
Bottom Skin Peak Match	27.40	–	–	**27.40 vs 70.33**	kN
Total Absorbed Energy	225.56	675.07	415.54	**387.70**	J
Energy Error	–	+199.3%	+84.2%	**+71.9%**	%
Sprague-Geers Phase (P)	–	26.57%	26.81%	**30.81%**	%
Sprague-Geers Magnitude (M)	–	219.20%	92.13%	**112.42%**	%
Sprague-Geers Comprehensive (C)	–	220.81%	95.95%	**116.57%**	%
Wallclock Execution Time	–	180.74 s	254.62 s	**490.30 s (8.2 min)**	s

- **Honeycomb Accordion Compaction ($2.5 \leq \delta \leq 10.0$ mm):** As the punch traverses the core space, the discrete foil cells buckle into progressive symmetric plastic folds. Foil self-contact (`/INTER/TYPE7/4`) prevents unphysical shell crossing without inducing artificial high-frequency energy spikes.
- **Bottom Skin Engagement and Aperture Shear ($10.0 \leq \delta \leq 15.2$ mm):** The compacted core plug contacts the inner face of the bottom laminate skin over the ∅30 mm die aperture. High membrane and shear stresses develop across the aperture perimeter, culminating in the peak puncture load of 70.33 kN at 15.19 mm, closely matching the physical stroke limit (15.80 mm).

6 Conclusion and Deliverables

The Phase 3 discrete meso-scale simulation successfully realizes:

1. Full discrete representation of 871 honeycomb cell walls with $5,226$ foil shells across the complete panel quadrant.
2. Rigorous characterization of the core-to-skin contact interface adhering to ASTM C273 and ASTM C297 standards with edge-to-surface (`/INTER/TYPE19`) formulation.
3. Highly optimized execution on 24 OpenMP threads, achieving 162,001 integration cycles in 8 minutes 10 seconds.
4. Generation and export of all 82 VTK state outputs for 3D visualization and validation.